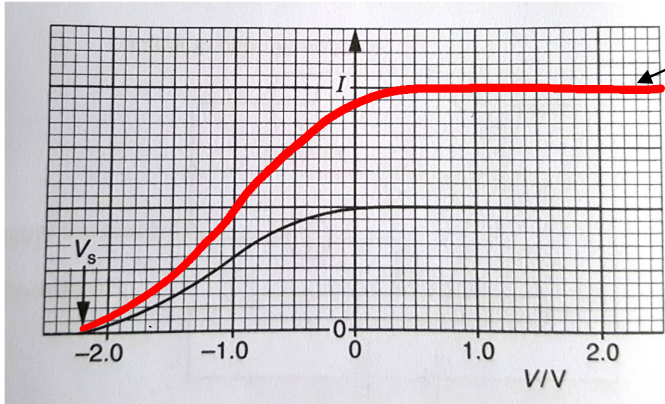


	Potential difference between C and D = $V_{CN} - V_{DN} = 8.0 - 4.32 = 3.68$ V
3 (a)	The loss in the electromagnetic radiation's individual photon energy must be greater than the work function of the metal. The difference between the loss in the individual photon energy and the work function of the metal is the maximum possible gain in the kinetic energy of a photoelectron which has absorbed the photon's energy, and hence, is emitted from the surface of M. Note: A large number of answers was in general terms such as threshold frequency rather than in terms of energy changes.
(b) (i)	If photoelectrons possess the maximum kinetic energy, equal to the gain in the electric potential energy eV_s, lose all their kinetic energy and just brought to rest at electrode C, all other photoelectrons with lower kinetic energy will lose their entire kinetic energy before they are able to reach electrode C. Note: A large number of answers did not answer the question, but simply explained the meaning of stopping potential.
(ii)	With positive values of V, all photoelectrons emitted from the electrode M are collected at electrode C. Since the electromagnetic radiation incident on M is of constant intensity, the rate of photoelectron emission from M is constant, independent of the potential difference V. Note: By merely stating that the potential has not effect on the current if the intensity of the radiation is constant is not answering the question.
(c)	Using $hf = eV_s + \phi$, From Fig. 3.2, $hf = 2.2 + 1.8 = 4.0$ eV $f = \frac{4.0 \times 1.60 \times 10^{-19}}{6.63 \times 10^{-34}}$ $= 9.653 \times 10^{14}$ $\approx 9.65 \times 10^{14} \text{ Hz}$
(d)	 Note that stopping potential is the same for same incident frequency.